

Scenario-Based Learning

A company works with a number of employees, and all the works are dependent on the employees. Even if one of the employees resigns, the assigned work may not be finished on time, so delivery of the project to the clients may be delayed. The company plans to make a solution for this. They want to know which employee may resign this month or next month. If they know this previously, they can arrange an alternative to avoid such problems. As an AI Engineer, you must give a solution to this.

- A) How will you achieve this in AI?
- B) Find out the 3-Stage of Problem Identification.
- C) Name the project.
- D) Create the dummy dataset.

A) How will you achieve this in AI?

Time-Series Based Supervised Binary Classification

Supervised Learning → Classification → using time-series/historical features

Artificial Intelligence



Machine Learning



Supervised Learning



Classification



Binary Classification



Time-Series / Historical Employee Data



Employee Attrition Prediction

B) Find out the 3-Stage of Problem Identification.

Stage 1: Identify the business problem — unexpected employee resignation may cause project delays.

Stage 2: Define the ML problem — use historical employee data to classify whether an employee is likely to resign in the next month (Yes/No).

Stage 3: Define the AI solution — develop a time-series-based supervised binary classification model that analyzes employee history and predicts potential resignations, allowing the company to plan replacements in advance.

C) Name the project:

AI-Based Employee Retention System

D) Create the dummy dataset:

Employee	Month	Satisfaction	Overtime	Absences	Salary	Resigned_Next_Month
E01	Jan	4	No	2	40000	0
E01	Feb	3	Yes	5	40000	0
E01	Mar	2	Yes	8	40000	1
E02	Jan	5	No	1	60000	0
E02	Feb	5	No	2	60000	0
E02	Mar	4	No	2	60000	0